MONISHA J

+91 7358417822 | monishajai2006@gmail.com | LinkedIn | Portfolio | Github

Education

B.E | Computer Science Engineering Sep 2023 - Present
Sri Sairam Engineering College, Chennai *CGPA:* 8.83*, No Standing Arrears*

Technical Skills

Core CS & Engineering	:	Data Structures & Algorithms, OOP, Design Patterns, Multithreading & Concurrency, SDLC, Test-Driven Development, Unit Testing
Languages	:	Python, Java, C, C++, JavaScript
Backend	:	Node.js, FastAPI, REST APIs, Computer Networks, DBMS, Operating Systems
Frontend	:	Next.js, React, Tailwind CSS, HTML5, CSS3
Databases	:	PostgreSQL, MySQL, MongoDB, Firebase
Tools & OS	:	Linux, Git, GitHub, AWS Cloud, Docker, Postman, Power BI

Projects

AI Honeypot System – Multi-Agent Backend & RAG Pipeline | *LangGraph, LangChain, FastAPI*

- **Cut manual threat analysis by 40%** by **designing** an autonomous **Agentic AI** multi-agent system in **LangGraph** with **MCP-based tooling** to simulate dynamic conversations and extract threat intelligence.
- **Indexed scam behavioral patterns across the full conversation dataset** by **constructing** a modular, **object-oriented RAG** pipeline using semantic **embeddings** and a **vector database**.
- **Automated intent classification and dataset labeling end-to-end** by **leveraging Claude Code** and custom agent **Skills**, following an iterative **SDLC** of design, build, and test cycles.

Property Pulse – ML Prediction Service | *FastAPI, Scikit-learn, XGBoost, Power BI*

- **Achieved 92% prediction accuracy across 5,000+ property listings** by **building** a supervised **ML pipeline** (Random Forest, XGBoost) with systematic feature engineering and k-fold cross-validation.
- **Isolated the key drivers of property valuation** by **conducting** exploratory data analysis and applying encoding, scaling, and dimensionality-reduction techniques to improve model generalization.
- **Enabled data-driven pricing decisions for 3+ stakeholder teams** by **developing** an interactive **Power BI** dashboard visualizing price distribution and prediction confidence.

Experience

Artificial Intelligence & Machine Learning Intern | *Infosys Limited* Sep 2025 - Oct 2025

- **Achieved 94% prediction accuracy on 10,000+ property records** by **building** a supervised regression pipeline (Random Forest + XGBoost ensemble) for real estate price prediction.
- **Improved model RMSE by 18% over baseline** by **performing** extensive feature engineering – encoding, outlier treatment, and correlation analysis – to identify key valuation drivers.
- **Shipped the model into production for stakeholder use** by **collaborating** with senior engineers to deploy it via **REST API** and validate outputs through interactive dashboards.

Database Management Intern | *Foxconn India Pvt Ltd* Jun 2025 - Jul 2025

- **Cut data retrieval latency by 30%** by **integrating SQLite** databases with manufacturing application systems for efficient storage and retrieval.
- **Reduced schema migration overhead by 25%** by **evaluating NoSQL** (MongoDB) solutions for flexible, large-scale manufacturing data.
- **Boosted operational visibility for 5+ stakeholder teams** by **collaborating** with the **MES** (Manufacturing Execution System) team to develop and test GUI dashboards for monitoring data pipelines.

Achievements & Publications

Competitions & Recognition

- **Finalist – AI Impact Summit Buildathon** (Feb 2026): Selected as a **top** 2% National Finalist among 40,000+ participants, **communicated** technical solutions to national-level stakeholders.
- **Winner – SDG Solvathon 4.0** (May 2025): First Place (Lead Presenter), Sri Sairam Engineering College, Chennai
- **AWS Nanodegree Program** (August 2026): Pursuing AWS Nanodegree in AI Programming, focusing on Python, Machine Learning, and AI development.

Publications

- **Aqua Pulse**: Advanced Sensor System for Water Health – Peer-reviewed and presented at ICCCT '25.
- **Property Pulse**: A Decision Support System for Predicting Property Value Trends – Presented at ICTMSD '26.